JAHANGIRNAGAR UNIVERSITY

Identification of Organic Compounds

B.Sc. (Hons.) in Chemistry

Department of Chemistry

Course No. Chem-335

Course teacher: Prof. Dr. Nasrin Jewena

Introduction

Most schemes for the identification of organic compounds begin with the classification of the purified unknown organic compounds on the basis of one or more of the followings:

(a) The elements present in the unknown organic compounds;
(b) The solubility of the unknown organic compounds in a particular solvent;
(c) The physical constants of the unknown organic compounds.

The next step consists in the application of the tests for functional groups; from the results it is possible to assign the unknown organic substances to a particular class of organic compounds. On the basis of all the data including the physical constants of the unknown organic compounds the most probable compound is selected from the literature. The final step consists in the preparation of two or three derivatives of the compound.

If the melting points of the respective known and unknown derivatives fall within **1.0** or **2.0** degree of each other, the identification is regarded as conclusive. An additional step, the determination of the mixed melting point is sometimes required. If a mixture of the known and the unknown derivatives melts within **1.0** degree of the melting point of either component alone, the identification is regarded as complete and conclusive.

Decomposition of organic compounds

Before the individual elements may be detected and identified, an organic compound must be decomposed with the production of ions or small molecules better capable of identification. For the purpose of decomposition the familiarly known ***"sodium fusion"*** method is generally used. Applying the method, the ***halogens***, ***nitrogen*** **and** ***sulphur*** may be detected. In this method, an organic compound is heated with metallic sodium using extensive decomposition, carbon is deposited and some **CO** and CO_2 are formed. Nitrogen, in the presence of carbon and sodium forms ***cyanide*** **(sodium cyanide, NaCN);** sulphur forms ***sulphide*** **(sodium sulphide,** Na_2S**)** and halogens form the corresponding ***halides*** **(sodium halides, NaX, X = Cl, Br, I).**
The residue obtained in the fusion is extracted with water.

Sodium fusion method

Add a small drop of the liquid compound, or a few mg of the solid compound to be tested to a clean, dry **4.0** inch pyrex test tube. Now add a piece of metallic sodium to the test tube, warm the bottom of the test tube slightly, and allow the mixture to stand for **2-3** minutes. Clamp the test tube in a vertical position and gradually apply heat to the bottom of the test tube until the sodium has melted. Being careful not to heat the sides of the hot test tube with the sample, drop again a few mg of the solid compound or a small drop of the liquid compound being tested directly onto the melted sodium. Heat the bottom of the test tube until the glass is **red hot**, and continue heating

for **2.0** minutes. Allow the test tube to room temperature and add **5.0** drops of methanol into the test tube.

If the residue is a globular mass, break it up with a clean glass rod. If gas bubbles are produced, wait until the reaction of metallic sodium with methanol is completed, then add **3.0** mL of distilled water. Boil the reaction mixture and filter it to remove the insoluble and floating materials. If at this points the solution is dark colored so as to obscure further tests it is probable that the amount of sample was too large or that the fused mass was too large or that the fused mass was not heated to a sufficiently high temperature. The fusion procedure should be repeated. The aqueous solution obtained by the above procedure is to be used in the tests for **N, S, X** [halogens, **Cl, Br** and **I**] and perhaps other elements.

$$\text{C, N, S, X} \xrightarrow{\text{Na, Heat}} \text{Na}_2\text{S, NaCN, NaCNS, NaX [X= Cl, Br and I]}$$

Detection of the elements in the sodium fusion filtrate

Detection of sulphur

(i) ***Lead acetate test*****:** About **0.20** mL of the alkaline filtrate or centrifugate obtained from the sodium fusion is placed in a **3.0** inch test tube**. 1.0** drop of lead acetate solution is added into test tube followed by dilute acetic acid until the solution is acidic. A **brown** to **black** precipitate or coloration indicates the formation of **lead sulphide (PbS)** and also indicates the presence of **sulphide (S^{-2}) ion.**

(ii) ***Sodium nitroprusside test*****:** Place about **0.20** mL of the filtrate from sodium fusion in a **3.0** inch test tube and add **1.0** drop of a **0.10%** solution of sodium nitroprusside. A **deep red** or **purple color** indicates the presence of the **sulphide** (S^{-2}) **ion.**

Detection of nitrogen

(i) ***Prussian blue test*****:** Place about **0.80** mL of the sodium fusion filtrate in a **4.0** inch test tube and add **1.0** drop of **3.0%** potassium fluoride **(KF)** solution**, 1.0** drop of **10.0% NaOH** solution and **15-20.0** mg of solid ferrous sulphate **($FeSO_4$)** crystals or solid ferrous ammonium sulphate crystals. Boil the mixture gently for **1.0** minute.

If a bluish colored precipitate does not exist at this point, add **2.0** drops more of the **10.0%NaOH** solution and reboil the mixture**.** Add a small droplet of **1.0%** ferric chloride **($FeCl_3$)** solution and again bring the mixture to boiling. Add dilute sulphuric acid **(dropwise and with shaking)** until the oxides of iron have just dissolved. Allow the test tube to stand for **2.0** to **5.0** minutes. A **blue color** or a **blue** precipitate is a positive test for nitrogen **(ferric ferro cyanide).**

A **green** or **greenish-blue coloration** indicates a weak test for nitrogen [perhaps due to poor fusion]**.** The addition of a drop of dilute phosphoric acid **(H_3PO_4)** to such a **greenish** mixture will usually cause the blue color to show up, since the yellow color is due to ferric chloride and the ferric ions are largely removed by the phosphoric acid. A **yellow** colored solution after acidification by the sulphuric acid indicates a negative test for **nitrogen.**

(ii) ***Copper acetate-benzidine test*****:** About **0.10** to **0.20** mL of the filtrate from sodium fusion is acidified with a drop of **10.0%** acetic acid and then **1.0** to **4.0** drops of copper acetate-benzidine reagent added carefully by means of a capillary pippette so that there is no appreciable mixing of the two solutions.

If **cyanide** ion is present a **blue ring** develops; if sulphide ion is present it is removed by adding **1.0** drop of lead acetate solution and centrifuging.

[The copper acetate-benzidine reagent is prepared from two stock solutions:
Solution **A:** containing **150.0** mg of benzidine in **100.0** mL of water and **1.0** mL of acetic acid; and solution **B:** containing **186.0** mg of copper acetate in **100.0** mL of water. The two solutions (solution **A** and **B**) are kept separately in two dark bottles and the reagent is prepared by mixing equal volumes of each solution (solution **A** and **B**) just before use].

Combine test for nitrogen and sulphur

In a **6.0** inch test tube take about **0.20** mL of the filtrate from sodium fusion and make it acidic with dilute **HCl.** A drop of freshly prepared **2.5% $FeCl_3$** solution is added into the acidic solution. A **blood red color** indicates the presence of **nitrogen** and **sulphur** altogether in the sodium fusion solution.

Detection of halogens

The presence of chlorine, bromine and iodine in an organic compound can be readily detected by the simple copper halide **(Beilstein's)** test, however, since a member of other organic compounds that do not contain halogen **[particularly those containing cyano and thiocynate functional groups]** give positive reactions to confirm the presence of halogens by means of one of the fusion tests. The alkaline filtrate from the sodium fusion mixture is first treated so as to eliminate the cyanide and sulphide ions **(if those are present)** and then tested by various procedures so as to detect the presence of **chloride, bromide, iodide or fluoride.**

(i) ***Beilstein's test for halogens*****:** A small loop is made on a piece of copper wire. The end of the copper wire having the loop is heated until no green color is detected, dipped (while hot) in some copper oxide, and reheated until the oxide gathers to the loop. Cool the copper wire. A minute amount of the compound to be tested is placed on the copper oxide (that is, dip the loop of the copper wire in a little of the unknown sample to be tested) and heated in the non luminous flame of a Bunsen burner, first in the inner, then in the outer zone near the lower edge. A **blue-green** flame indicates the presence of **chlorine, bromine, or iodine**; the color is due to the vapor of **copper halide**. Copper fluoride is not volatile and hence the test is not suitable for the detection of **fluorine.**

A number of compounds that contain no halogen, such as urea, thiourea, hydroxyamines, pyrimidines, pyridine derivatives, quinoline, copper cyanide and some nitrogen containing carboxylic acids give strong positive tests. Another drawback of this test is the possibility of highly volatile liquids evaporating completely prior to the wire becoming sufficiently hot to cause decomposition, thus resulting in possible false-negative results. It is best to confirm a positive test with the sodium fusion test.

(ii) ***Silver halide test*****:** The sodium fusion filtrate may contain one or more of the following ions: **fluoride, chloride, bromide and iodide.** Silver fluoride is highly soluble in water; hence the presence of fluoride ions cannot be detected by the addition of silver ions. Silver cyanide and silver sulphide are both sparingly soluble in water; hence, if either nitrogen or sulphur has been found present by previous tests, the **cyanide or sulphide ions** must be removed before adding silver ions to test for the halides.

To remove cyanide or sulphide ions add about **1.0** mL of distilled water to **0.30** to **0.50** mL of the sodium fusion filtrate in a **4.0** inch test-tube and acidify the solution by addition, dropwise, concentrated nitric acid **(HNO_3).** Hold the tube under the hood and boil the mixture until the volume has been reduced by one half. Cool the mixture and add **3.0** drops of **5.0%** silver nitrate

($AgNO_3$) solution. A **white** or slightly **yellow** precipitate indicates the presence of ***chloride, bromide or iodide*** ions. If neither nitrogen nor sulphur has been found present, the fusion filtrate may be acidified with nitric acid and tested without boiling.

Differentiation between chloride, bromide and iodide in the fusion extract

Numerous methods have been suggested for the detection of iodine, bromine and chlorine, either alone or in mixtures, based on the differences in the case of oxidizability of their ions to the elements. **The following procedures are generally applied:**

(i) ***Reaction with nitric acid***: Place about **0.50** mL of the sodium fusion filtrate into a **4.0** inch test tube. Add **0.50** mL (**8.0** drops) of carbon tetrachloride (CCl_4) and about **1.0** mL (**15-20.0** drops) of concentrated nitric acid (HNO_3). Shake the test tube and allow the liquid of the test tube to stratify and observed the colors in the carbon tetrachloride layer:

* **Iodine** will produce **violet color** in the CCl_4 layer;
* **Bromine** will produce **deep brown** or **tan** or **tan red** color in the CCl_4 layer; and
* **Chlorine** will produce **no color** in the CCl_4 layer.

(ii) ***Reaction with chlorine-water and hydrogen peroxide***: About **0.20** mL of the sodium fusion filtrate is placed in a micro tube with **1.0** to **2.0** drops of chloroform and **1.0** drop of dilute sulphuric acid. A drop of freshly prepared chlorine-water solution or **2.0** to **3.0** drops of **3.0%** hydrogen peroxide solution is added into the micro tube and the mixture is shaken and allows the liquid of the micro tube to stratify and observed the colors in the chloroform layer:

* **Iodine** will produce **violet color** in the $CHCl_3$ layer;
* **Bromine** will produce **deep brown** or **tan** or **tan red** color in the $CHCl_3$ layer;
* **Chlorine** will produce **no color** in the $CHCl_3$ layer.

Detection of individual halogen in a halogen mixture

(i) Place about **0.50** mL of the sodium fusion filtrate into a **4.0** inch test tube. Add about **0.50** mL (**8.0 drops**) of carbon tetrachloride (CCl_4) and about 1.0 mL (**15-20** drops) of concentrated nitric acid (HNO_3). Shake the test tube and allow the liquid of the test tube to stratify and observed the colors in the carbon tetrachloride layer:
The presence of **iodine** is indicated by a **violet color** in the CCl_4 layer.

If iodine is present, remove the carbon tetrachloride layer by a pipette and add **0.50** mL of carbon tetrachloride to the same original test solution. Add **1.0** drop of concentrated nitric acid, shake the reaction mixture and allow it to stratify. If the **violet** color still shows, remove the lower layer by means of a pipette and without adding more nitric acid repeat the extraction of the iodine by carbon tetrachloride until the lower layer remains essentially colorless. Now add **2.0** mL of concentrated nitric acid to the reaction mixture being tested. Shake the solution and allow to stratify. The presence of **bromine** is indicated by a **tan** or **tan-red color** in the carbon tetrachloride layer. Repeat the extraction of the bromine, if present, until the lower layer becomes colorless. Add about **3.0** drops of **5.0%** silver nitrate solution to the aqueous layer. The immediate formation of a **white** precipitate indicates the presence of **chlorine**.

Classification of organic compounds by solubility

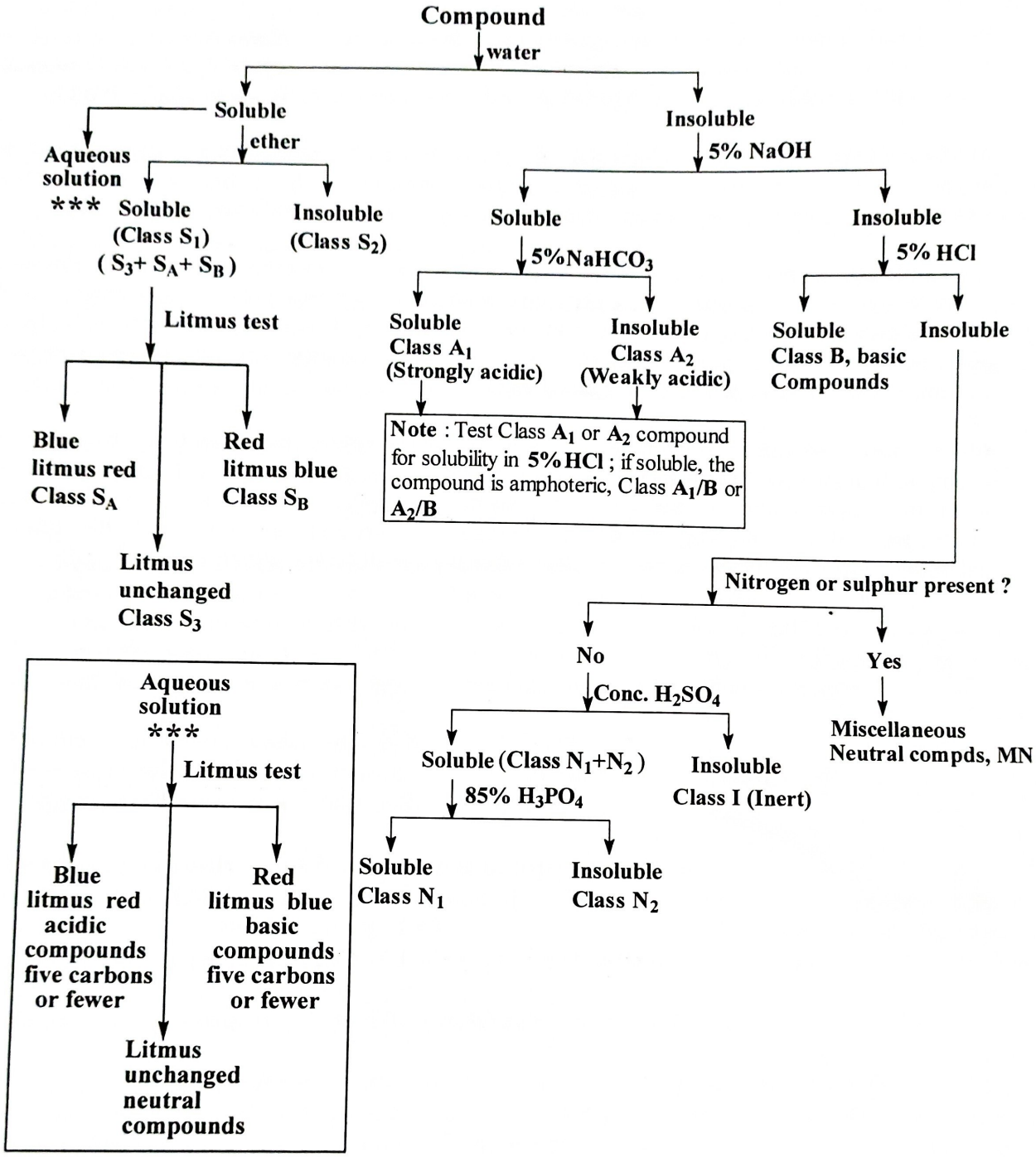
After analysis for the elements present, the next important step is the determination of solubility of the unknown organic compounds to be tested. It provides with many informations as regards the nature of the compounds. These informations are very much helpful prior to the specific test for functional groups.

Three general kinds of information can often be obtained about an unknown organic substance by a study of its solubility behavior in various solvents: namely, **water, ether, 5.0%NaOH solution, 5.0%$NaHCO_3$ solution, 5.0%HCl solution, cold concentrated H_2SO_4 and 85.0% H_3PO_4.**

(i) The presence of a functional group is often indicated. For instance, because hydrocarbons are insoluble in water, the mere fact that an unknown compound (such as diethyl ether) is partially soluble in water indicates that a polar functional group is present in the unknown molecule.

(ii) Solubility in certain solvents often leads to more specific information about the functional group. For example, benzoic acid is insoluble in polar solvent, water, but is converted by **5.0% sodium hydroxide** solution to a salt, sodium benzoate, which is readily water soluble. In this case, then, the solubility in **5.0%NaOH** solution of a water insoluble unknown organic compound is a strong indication of an acidic functional group.

(iii) Certain deductions about molecular size and composition may sometimes be made. For example, in many homologous series of monofunctional compounds, the members with less than about five carbon atoms are water soluble, whereas the higher homologs are insoluble in water. On the basis of the foregoing discussions the organic compounds are divided into the following solubility classes depending on their solubility behavior in different solvents:



Solubility classifications of organic compounds

Class S_2

Only C, H, O present: Dibasic and polybasic acids, hydroxy acids, polyhydroxy alcohols, polyhydroxy phenols, carbohydrates.

Nitrogen present: Salts of organic acids (RCOONa, RSO_3Na), amine hydrochlorides (RNH_3Cl), amino acids.

Halogen present: Haloacids.

Sulphur present: Sulphonic acids.

Nitrogen & sulphur present: Amino sulphonic acids.

Class S_3
Only C, H, O present: Monofunctional alcohols, aldehydes, ketones and esters with five carbon atoms or fewer.
Nitrogen present: Monofunctional amides, nitriles with five carbons or fewer.
Halogen present: Halogen substituted **C, H, O,** compounds.

Class S_A
Monofunctional carboxylic acids with five carbon atoms or fewer.

Class S_B
Monofunctional amines with six carbon atoms or fewer.

Class A_1
Only C, H, O present: Strong organic acids: carboxylic acids with more than six carbon atoms; anhydrides, phenols with electron withdrawing groups in the ***ortho-*** and ***para-*** positions; β- diketones (1,3-diketones).
Nitrogen present: Nitro acids.
Halogen present: Haloacids, acid halides.
Sulphur present: Sulphonic acids.
Sulphur &;oikliikj halogen present: Sulphonyl halides.

Class A_2
Only C, H, O present: Weak organic acids**:** phenols, enols, oximes, imides, sulfonamides, thiophenols, all with more than five carbon atoms; β- diketones (1,3-diketones); nitro compounds with α-hydrogens.
Nitrogen present: Amino acids, nitrophenols.
Halogen present: Halo phenols.

Class B
Aliphatic amines with more than eight or more carbon atoms; aromatic amines such as anilines (only one phenyl group attached with nitrogen); some ethers.

Class MN
Miscellaneous neutral compounds containing nitrogen or sulphur and having more than five carbon atoms.
Nitrogen present: Anilides, toluidides, amides, imides, nitro arylamines, nitro-hydrocarbons.

Class N_1
Alcohols, aldehydes, methyl ketones, cyclick ketones, esters with one functional groups and more than five but fewer than nine carbon atoms, etc.

Class N_2
Unsaturated hydrocarbons: alkenes, alkynes; some aromatic compounds (especially those with activating groups), ethers, ketones (other than those cited in class N_1).

Class I
Saturated hydrocarbons, alkyl halides, arylhalides, other deactivated aromatic compounds, diaryl ethers, etc.

Solubility determinations

A compound is considered soluble in a solvent if **30.0** mg of it dissolves in **1.0** mL of the solvent at room temperature. The solvents to be used or the solubility classifications are in the order in

which they should be used; **water, ether, 5.0%HCl solution, 5.0%NaOH solution, 5.0%$NaHCO_3$ solution, concentrated H_2SO_4 and 85.0% H_3PO_4.** All solubilities are to be determined at room temperature. It is permissible to warm the mixture slightly during the solubility test but the mixture should be cooled before a decision is reached as to its solubility.

If the compound is soluble in water its solubility may be tested in ether, but in no other solvents. If the compound is insoluble in water, it needs not be tested with ether. Water insoluble compounds that are soluble in **5.0% sodium hydroxide** solution should also be tested with **5.0% sodium bicarbonate** solution to distinguish between relatively strong acids and the weakly acidic compounds. If an obvious chemical reaction occurs in any solvent, the compound may be considered as soluble in that solvent.

A few of the amine hydrochlorides are slightly soluble in **5.0%HCl;** hence in these cases the original amine will react when it is first added to the aqueous acid to form the amine hydrochloride and will then precipitate as that salt. To determine whether the insoluble material is the original amine or a salt that has been formed, a mixed melting point determination should be made, using the washed and dried solid from the acid solution with some of the original sample. If the compound contains **N** or **S** and is insoluble in **water, 5% HCl** solution, **5%NaOH** solution, it is placed in a special miscellaneous class without a test for its solubility in concentrated H_2SO_4. In some cases other concentration of **HCl, NaOH** and **$NaHCO_3$** solutions are used. If there is doubt as to whether or no appreciable amount of a substance has dissolved in **5%HCl, 5%NaOH** or **5%$NaHCO_3$** reagents, the clear solution should be separated from the excess of the compound and neutralized to see if the original compound will separate.

Functional groups determination

Detection of amino groups

Classification tests of the amines

(i) ***Quinhydrone test*:** Place **6.0** mL of water in a **6.0** inch test tube and add **15.0** to **20.0** mg **(or 1.0 drop)** of the amine. Shake the mixture. If the amine dissolves in water, add **6.0** mL more of water. If the substance did not dissolve in the first **6.0** mL of water, add **6.0** mL of ethanol. Shake the mixture. To the test tube, now containing **12.0** mL of solvent, add **1.0** drop of a **2.5%** solution of quinhydrone in methanol. Shake the mixture and allow it to stand for **2.0** minutes and observe the colors.

Aliphatic amines produce colors with only one drop of the reagent;
***Primary amines produce a violet color;**
***Secondary amines produce a rose color; and**
***Tertiary amines produce a yellow color.**

Aryl amines generally do not produce any color with only **1.0** drop of the reagent, hence if no color develops from **1.0** drop; add **5.0** more drops of the quinhydrone reagent. Shake the mixture and allow it to stand for **2.0** minutes and observe the colors.
Aryl amines produce the following colors;
***Primary amines produce rose color;**
***Secondary amines produce amber color; and**
***Tertiary amines produce yellow color.**

(ii) ***Hinsberg test*:** Add **2.0** drops of the amine to about **2.0** mL of pyridine; then add about **0.80** mL of freshly prepared **2.0%** aqueous **NaOH** solution. Shake the mixture thoroughly and add **1.0** drop of benzenesulphonyl chloride. Shake the mixture again and observe the colors:
***A yellow color indicates a primary amine;**

***An orange color indicates a secondary amine; and**
***Deep red or purple color indicates a tertiary amine.**

Test for primary aliphatic amines

(i) ***Sodium nitroprusside test*****:** To about **0.30** mg of the compound taken in a test tube add about **5.0** mL of water, about **1.0** mL of acetone and a few drops of **1.0%** sodium nitroprusside solution. Allow the mixture to stand for **2.0** minutes. Formation of a **violet-red** color indicates the presence of aliphatic primary amine.

(ii) ***Carbylamine (isocyanide) test*****:** Boil a little quantity of the compound with **2.0** drops of chloroform and about **2.0** mL of alcoholic **KOH** solution. An intolerable offensive odor of a carbylamine indicates the presence of a primary aliphatic amine.

(iii) ***Nitrous acid test*****:** Dissolve a little quantity of the compound in dilute **HCl** and cool (at **0-5°C**). Now add **10.0%** aqueous **$NaNO_2$** solution (which is previously chilled at **0-5°C**). Formation of **brisk effervescence** indicates the presence of aliphatic primary amine.

Test for primary arylamine

(i) ***Diazotization test*****:** In one **4.0** inch test tube, mix about **30.0** mg of the amine with about **1.0** mL of water and **4.0** drops of concentrated **H_2SO_4.** Cool the test tube in a chilling bath (at **0-5°C**). Add about **1.0** mL of **10.0% $NaNO_2$** solution **(which is previously chilled at 0- 5°C)** dropwise and with shaking. Now add dropwise, the previously chilled (at **0-5°C**) sodium napthoxide solution. **[solution of 100.0 mg of *β*-naphthol in 2.0 mL of 10.0%NaOH solution].** Formation of a **red or orange dye** indicates the presence of **primary aryl amine.**

Note: 2.4-dinitroaniline does not diazotize. *o*-diamines do not diazotize, but form dark colored azoimides. *m*-Diamines diazotize and couple with undiazotized amine to from brown dyes.

(ii) ***Carbylamine (isocyanide) test*****:** Boil a little quantity of the compound with **2.0** drops of chloroform **($CHCl_3$)** and about **2.0** mL of alcoholic **KOH** solution. An intolerable offensive odor of a carbylamine indicates the presence of a **primary aryl amine**.

Tests for secondary amines

(i) ***Nitrous acid test*****:** Prepare a solution of nitrous acid by adding about **1.0** mL of ice cold (at **0-5°C**) concentrated **HCl** to a about **2.0** mL **20.0%** aqueous **$NaNO_2$** solution. Add gradually this solution to about **0.20** gm of the compound in **10.0** mL dilute **HCl**. An oily dark colored liquid indicates the presence of **secondary amines.**

(ii) ***Libermann's nitroso reaction*****:** Prepare a solution of nitrous acid by adding about **1.0** mL of ice cold (at **0-5°C**) concentrated **HCl** to a about **2.0** mL of **20.0%** aqueous **$NaNO_2$** solution Add gradually this solution to about **0.20** gm of the compound in **10.0** mL dilute **HCl**. The reaction mixture is then heated with **1-2** drops each of phenol and concentrated **H_2SO_4.** A **blue** coloration which changes to **red** on dilution with water and **blue black** with aqueous alkali **(5.0%NaOH)** confirms the presence of **secondary amines.**

Specific test for secondary aliphatic amines

Sodium nitroprusside test**:** Add about **1.0** mL of a freshly prepared **5.0%** solution of acetaldehyde to about **5.0** mL of a dilute aqueous solution of the amine **(1.0 or 2.0 drops of amine in about 5.0 mL of water)**. Add **1.0** or **2.0** drops of **10.0%** sodium nitroprusside solution

and **2.0** drops of dilute **$NaHCO_3$** solution. A **blue** color will develop within **3.0** minutes. On standing the solution may change color to **green** and finally to **yellow.**

Specific test for diphenyl amine

Dissolve about **30.0** to **40.0** mg of the compound in about **2.0** to **4.0** mL of concentrated sulphuric acid **(H_2SO_4)** or hydrochloric acid **(HCl)** and add **1-2** drops of dilute nitric acid **(HNO_3)** or sodium nitrite **($NaNO_2$)** solution. A **deep blue** color indicates the presence of **diphenyl amine.**

Test for tertiary amines

(i) ***Ohkuma test*****:** Add **5-6.0** drops of the citric acid-acetic anhydride reagent to the alcoholic solution of the compound shake and heat in a boiling water bath. Development of a **red** or **purple** color within **2-3.0** minutes indicates the presence of tertiary amine. **[Preparation of citric acid-acetic anhydride reagent: Add 2.0 gm of citric acid to 100.0 mL of acetic anhydride, shake and heat to get a clear solution].**

(ii) ***Nitrous acid test*****:** Prepare a solution of **HNO_2** by adding ice cold (at **0-5°C**) dilute **HCl** to **1.0%** aqueous **$NaNO_2$** solution. Add gradually this solution to about **0.20** gm of the compound in **10.0** mL of dilute **HCl.**

***No reaction indicates aliphatic tertiary amine;**

***Production of green and brown colored salt indicates the presence of aromatic tertiary amines.**

Test for nitro compounds

(i) ***Diphenylamine test*****:** Place **1.0** or **2.0** drops of an ether or benzene solution of the compound in a small test tube. Add **1.0** or **2.0** drops of a **5.0%** solution of diphenylamine in benzene. Hold the test tube in boiling water to evaporate off the solvents and to melt the mixture. Colored molecular compounds are formed between molten diphenylamine and nitro compounds. The color appears on heating the mixture to **100°C** or less and fades the color when the melt is cooled but reappears the color on reheating.

(ii) ***Test with griess reagent*****:** Add a few milligrams of the substance to be tested to a 6.0 inch test tube. Cover the top of the test tube with a filter paper that has been moistened with griess reagent. Heat the test tube for **1.0** or **2.0** minutes. A **pink** to **red** coloration on the filter paper is a positive test.**[Griess reagent is freshly prepared by using equal volume of 1.0% sulphanilic acid solution and 0.10% 1-naphthyl amine].**

(iii) Reduction of nitro compound followed by diazotization and then by coupling

Reduction of nitro compound**:** In a **6.0** inch test tube place about **20.0** mg of the nitro compound to be tested and about **10.0** mg of tin. Add about **1.0** mL of concentrated **HCl** dropwise. When the vigorous reaction subsides heat to boiling with care for **10.0** minutes, cool the reaction mixture and dilute with about **2.0** mL of water.

Diazotization**:** In a **4.0** inch test tube take **1-2.0** mL of the above dilute reaction product and **1-2.0** drops of concentrated **HCl**. Cool the reaction mixture in a chilling bath (at **0-5°C).** Add **1.0** mL of **10.0%** solution of **$NaNO_2$ (which is previously chilled at 0-5°C)** dropwise and with shaking. Now add dropwise the previously chilled (at **0-5°C**) sodium naphthoxide solution **(solution of 100.0 mg of β-naphthol in 2.0 mL of 10.0% NaOH).** Formation of a **red** or **orange dye**

indicates the presence of **primary aryl amine.** This test indicates that the sample was an aromatic nitro compound that was reduced and formed primary aromatic amine.

Special test for di-and trinitro benzene

In a test tube place a little quantity of the sample, about **5.0** mL of acetone and add with shaking about **2-3.0** mL of **5.0%** aqueous **NaOH** solution. Note the color produced:

***Dinitro benzene: purple-blue color;**
***Trinitro benzene: deep red color.**

General test for amide

(i) ***Nitrous acid test*:** To about **0.10** g of substance add about **2.0** mL of dilute **HCl** followed by **2.0%** aqueous $NaNO_2$ solution **(which is previously chilled at 0-5°C)** and shake. Brisk effervescence due to evolution of nitrogen gas indicates the presence of **an amide**.

(ii) ***Sodium hydroxide-copper sulphate test*:** Place about **50.0** mg of the substance in a **4.0** inch test tube and add about **2.0** mL of **20.0% NaOH** solution. Evolution of NH_3 gas indicates the presence of amide or imide. Place a small filter paper over the top of the test tube. Add **2.0** drops of aqueous **10.0%**$CuSO_4$ solution on the filter paper covering the test tube. Heat the test tube for **1.0** minute. A **blue color** will develop on the filter paper if the substance is an **amide.**

Specific test for aliphatic amide

***Hydroxamate test*:** Mix about **50.0** to **60.0** mg of the substance with about **30.0** mg of solid hydroxylamine hydrochloride in a **4.0** inch test tube and add about **2.0** mL of ethanol. Boil the reaction mixture on water bath for **5.0** minutes. Cool and add few drops of aqueous **10.0%** $FeCl_3$ solution in methanol. Formation of **blue-red color** indicates the presence of **aliphatic amide.**

Specific test for aromatic amide

***Ferric chloride-hydrogen peroxide test*:** To the aqueous solution (about **50.0** mg in **3.0** mL H_2O) of the sample add **7-8** drops of **6.0%** H_2O_2 and heat the mixture to near boiling. Cool the solution and add **1-2.0** drops of **5.0%** $FeCl_3$ solution. Formation of **bluish-red** color in cold within two minutes or a brown color on warming indicates the presence of an aromatic amide. The addition of a few mL of **10.0% NaOH** to the final mixtures produces a clear **deep reddish-brown** solution.

Test for anilide

(i) ***Tafel's test*:** Add about **10.0** mg of the sample to be tested to **6.0** drops of concentrated H_2SO_4 in a dry test tube. Stopper the test tube and shake it vigorously and then add a little of finely powdered $K_2Cr_2O_7$. A **bluish pink color** or **red** or **violet color** changing to **green** on heating confirms the presence of an anilide.

(ii) ***Diazotization test*:** Boil about **0.10** g of the sample to be tested with about **5.0** mL of dilute **HCl**, cool and filter. Treat the filtrate with **4.0** mL of aqueous $NaNO_2$ solution on an ice bath (at **0-5°C**) and then pour all this content into a beaker (ice-bath at **0-5°C**) containing **10.0%** alkaline **β-napththol.** Formation of a **bright orange red** dye indicates the presence of **anilide**.

(iii) ***Carbylamine test*:** Carbylamine test offering offensive vapours is also a positive test for anilide. Boil a little quantity of the compound with **2.0** drops of chloroform **($CHCl_3$)** and about **2.0** mL of alcoholic **KOH** solution. An intolerable offensive odor of a carbylamine indicates the presence of anilide.

Test for urea and substituted ureas

(i) ***Biuret test*****:** Heat gently a small quantity of the sample to be tested in a dry test tube for **2-3** minutes. NH_3 gas evolves and the residue solidifies with the formation of Biuret **($NH_2CONHCONH_2$).** Dissolve the residue obtained in **10.0% NaOH** solution. Warm and cool. Now add **2.0** drops of very dilute $CuSO_4$ solution. Development of a **violet color** confirms the presence of urea or substituted urea.

(ii) ***Sodium hypobromite test*****:** Take small amount of urea in aqueous solution and add to it sodium hypobromite solution, brisk effervescence will be evolved which indicates the presence of urea or substituted urea.

Test for thiourea

(i) ***$FeCl_3$ test*****:** In a dry test tube heat a small quantity of the sample to be tested. Dissolve the fused mass in water and add few drops of **10.0% $FeCl_3$** solution. Formation of **blood red** color indicates the presence of thiourea or mono substituted thiourea.

(ii) ***Lead acetate test*****:** In a test tube heat a small quantity of the sample with about 2.0 mL of aqueous **KOH** solution. Now cool the solution and add one drop of aqueous lead acetate solution. Dark **brown** or **black** color of **PbS** is a positive test for **thioureas**.

Test for carbonyl compounds

Test for aldehydes or ketones (General test for carbonyl groups)**:** Place about **0.50** mL of a saturated solution of **2,4-dinitrophenyl hydrazine** in **2N** hydrochloric acid in a **6.0** inch test tube. Add to it a solution of about **30.0** to **40.0** mg of the compound in about **0.50** mL of methanol **[if it is slightly soluble in water, it may be necessary to add more alcohol than that recommended]**. Stopper the test tube and shake it vigorously. If a precipitate does not form heat the mixture to boiling for **30.0** seconds and shake it again. The formation of a color precipitate of **2,4-dinitrophenyl hydrazone** indicates the positive test for carbonyl group. Most aromatic carbonyl compounds produce **red products**, whereas aliphatic carbonyl compounds produce **yellow products.**

Test for aldehydes only

(i) ***Reaction with Schiff's reagent*****:** Add **3.0** drops of an aldehyde (or about **0.50** mg of aldehyde if the aldehyde is solid) to be tested to about **2.0** mL of colorless Schiff's reagent. A **wine-purple** color will develop within **10.0** minutes which indicates the presence of aldehydes.

(ii) ***Methone test*****:** Add about **50.0** mg of the aldehyde to be tested to **1.0** mL of water and then add **3.0** drops of a **5.0%** solution of methone in ethanol. Shake the mixture. The formation of a **milky suspension** within **2.0** minutes is a positive test for aldehydes.

(iii) Oxidation test

(a) ***Reaction with Tollen's reagent*****:** Add about **30.0** mg of the compound to be tested to **2.0** mL of freshly prepared ***Tollen's reagent***, shake the test tube and allow it to stand for **10.0** minutes. If no reaction occurs in this time, place the test tube in a beaker of water at about **35°C** temperatures for **5.0** minutes. The formation of a **silver mirror** is a positive test for aldehydes.

(b) ***Reaction with Benedict's reagent*****:** Only aliphatic aldehydes **(but not aromatic aldehydes)** produce a **yellow** to **orange** colored precipitate or suspension when heated with Benedict's reagent. To a solution or suspension of about **30.0** mg (or about **1.0** mL, if the sample is liquid) of the compound to be tested in **2.0** mL of water, add about **2.0** mL of the freshly prepared ***Benedict's reagent*** and shake the test tube. The formation of **yellow** to **orange** precipitate or

suspension of copper (I) oxide indicates the presence of aldehydes. If no colored precipitate or suspension of copper (I) oxide is formed in this time, heat the reaction mixture to boiling in a boiling water bath and cool. Note whether any colored solid is formed. The formation of a **yellow** to **orange** colored precipitate of copper (I) oxide is a positive test for aldehydes.

Tests for methyl ketones only

***Sodium nitroprusside test*:** Compounds which give the positive test for carbonyl group but give negative tests for aldehydes are considered as ketones. The ketones having a methyl or methylene group attached to the carbonyl group give the following test: Mix a drop of the aqueous or alcoholic solution of the compound to be tested with **1.0** drop of **5.0%** sodium nitroprusside solution and **1.0** drop of **30.0%NaOH** solution. A slight **red** or **yellow color** generally develops within a few minutes. Add **1.0** or **2.0** drops of glacial acetic acid. A **red** or **blue color** on the addition of the acid is a positive test for ketones having methyl or methylene group attached to the carbonyl group.

Test for phenols

Phenols are to be suspected when a compound falls in solubility division A_2. Generally more than one test for phenols must be made before a conclusion can be drawn.

(i) ***Ferric chloride test*:** Dissolve about **30.0** to **50.0** mg of the compound to be tested in **1.0** to **2.0** mL of water, or a mixture of water and alcohol, and add upto **3.0** drops of a **2.5%** aqueous solution of ferric chloride. Phenol, resorcinol, ***m*-$C_6H_4(HO)_2$; *ortho*-, *meta*-** and ***para*-**cresol give violet or blue color. Hydroquinone, ***p*-$C_6H_4(HO)_2$** undergoes oxidation; it gives a deep green solution (crystals may separate), and on further addition of ferric chloride, a yellow solution of *p*-benzoquinone is obtained. α-and β-Naphthol do not give characteristic coloration with aqueous solution of ferric chloride. **[Most** nitrophenols and ***para*-**substituted phenols do not give this ferric chloride test, **-CHO**, **-COOH** and **-$COCH_3$** groups on the aromatic ring also prevent the reaction of phenols with ferric chloride solution**].**

(ii) ***Nitrous acid test:*** Add about **50.0** mg of the compound to **1.0** mL of concentrated **H_2SO_4** in a **4.0** inch test tube and then add about **20.0** mg of solid **$NaNO_2$.** Shake the test tube and warm it slightly. A positive test is indicated by a **green**, **blue**, or **purple** color. Cautiously pour the mixture into **5.0** mL of water. The color will generally change to **red** or **blue-red.** Make the solution alkaline by adding **20.0 % NaOH** solution. The color in alkaline solution is generally **blue** or **green.**

(iii) Solubility test

(a) Phenols are soluble in **5.0%** aqueous **NaOH** solution but insoluble in **5.0%** aqueous **$NaHCO_3$** solution.

(b) Phenols containing electron withdrawing groups (such as picric acid, **2,4**-dinitroptnol etc.) dissolved in **5.0%** aqueous **$NaHCO_3$** solution giving bubbles of **CO_2.**

(iv) ***Bromine-water test*:** Dissolved about **0.10** mg of the compound in **10.0** mL of water and add bromine-water until the color of the latter persists. A **white** precipitate will form if a phenol is present.

Tests for nitro phenols

(i) ***Reaction with sodium hydroxide solution*:** Add about **30.0** mg of the compound to **2.0** mL of a **10.0%** aqueous solution of **NaOH.** Note the color produced:

***p*-Nitrophenol: Yellow color;**

***o*-Nitrophenol: Yellow-orange color;**
***2,4-Dinitrophenol: Bright yellow color.**

Test for alcohols

(i) ***Vanadium oxine test*:** Place about **30.0** mg of the compound to be tested in a **4.0** inch test tube and add about **0.50** mL of a solution of ammonium vanadate. Now add **1.0** or **2.0** drops of a **2.5%** solution of **8**-hydroxyquinoline **(oxine)** in **6.0%** acetic acid. A **red** complex is produced. Solution of the red complex in **0.50** to **0.70** mL of benzene becomes **pink** to **red** showing the presence of alcohol.

(ii) ***Xanthate test*:** Add **1.0** pellet of solid **KOH** to **0.50** mL of alcohol in a dry test tube and heat until the **KOH** dissolve. Cool the test tube and add **1.0** mL of ether. Add dropwise carbon disulphide until a pale **yellow** precipitate forms. If a precipitate does not form by the time **0.50** mL of carbon disulphide has been added, the test should be considered as negative.

Tests to distinguish between primary, secondary and tertiary alcohols

(i) ***Reaction with N-bromosuccinimide*:** Dissolve about **50.0** mg of the alcohol in **1.0** mL of a **0.01%** solution of bromine in CCl_4 in a **4.0** inch test tube. Add **20.0** mg of N-bromosuccinimide and place the test tube in a water bath at **78** to **80°C** temperature. At this temperature the solution will boil gently. Any color change from the initial **pale yellow** will occur within **13.0** minutes.

***Primary alcohols** give a permanent **orange color**; and
***Secondary alcohols** produce a transitory **orange color** which fades **(usually very rapidly),** leaving the solution colorless after continued boiling. After the tube is cooled an **orange** precipitate is often apparent in the case of a secondary alcohol do not produce color changes with the reagent.

(ii) ***Lucas reagent test*:** Add **2.0** or **4.0** drops of the alcohol to be tested to about **2.0** mL of Lucas reagent **($ZnCl_2$ dissolved in concentrated HCl)** in a **3.0** inch test and shake the mixture vigorously. Allow the mixture to stand at room temperature. A reaction is detected by the cloudiness of the solution, owing to the formation of an insoluble **(milky solution)** alkyl chloride.

***Tertiary alcohols** react immediately;
***Secondary alcohols** react within **2.0** to **3.0** minutes; and
***Primary alcohols** require a much longer time. If there is any doubt as to whether the alcohol is 2^o or 3^o, mix **2.0** to **4.0** drops of the alcohol with only about **2.0** mL of concentrated **HCl**. **Secondary** alcohols are not converted to alkyl chlorides under these conditions, whereas 3^o **alcohols** yield the insoluble milky alkyl chloride within **10.0** minutes.

Test for carboxylic acids

(i) ***Solubility test*:** Carboxylic acids are soluble in **5.0 %** aqueous **NaOH** and in **5.0%** aqueous $NaHCO_3$ solution [the later reaction is accompanied by the evolution of CO_2 gas].

(ii) ***Bromine test*:** Non-reactive towards bromine-water or a solution of bromine in carbon tetrachloride (unless an ethylene or acetylenic or unsaturated group is also present in the acid molecule to be tested).

(iii) ***Conversion of carboxylic acid to anilide*:** Place about **100.0** mg (or about **3.0 - 4.0** drops, if the sample is liquid) of the carboxylic acid in a **6.0** inch test tube and add **8.0** drops of thionyl chloride. Insert a micro condenser and gently reflux the materials for **10.0** to **20.0** minutes. Cool the reaction mixture, and add a solution of about **100.0** mg of aniline in **3.0** mL of benzene, and

heat the mixture in a hot water-bath or on a steam bath for **2.0** minutes. Add about **2.0** mL of cold **5.0%** aqueous **NaOH** solution and **2.0** mL of ice-cold water into the reaction mixture and shake the test tube vigorously. The formation of a white precipitate of **anilide** indicates the presence of **carboxylic acids [Benzene is a known carcinogen. Use benzene in the hood, do not breathe the benzene vapors, and avoid the contact of benzene with the skin].**

(iv) ***Conversion of carboxylic acid to acid chloride*:** Place about **100.0** mg (or **3.0- 4.0** drops, if the sample is liquid) of the carboxylic acid in a **6.0** inch test tube and add **8.0** drops of thionyl chloride. Insert a micro condenser and gently reflux the materials for **10.0** to **20.0** minutes. Add **2.0-3.0** drops of benzyl amine into the reaction mixture. To this reaction mixture finally add **2.0** mL of cold water and shake the test tube vigorously. The formation of a white precipitate of N-benzyl amide which is slightly soluble in water indicates the presence of carboxylic acids (**perform the entire procedure in the hood**).

Tests for anhydrides and acid halides

***Ferric hydroxamate test*:** Add about **30.0** to **40.0** mg of the compound to about **0.50** mL of a **1.0N** solution of hydroxyl ammonium chloride in alcohol. Add **2.0** drops of **6M HCl** solution to the reaction mixture. Warm it slightly for **2.0** minutes and then boil for a few seconds and cool the solution. Add **1.0** drop of **10.0% $FeCl_3$** solution. The formation of **reddish-blue** or **bluish-red color** is a positive test for acid anhydrides or acid halides. If the color is more red than blue, adjust the **pH** of the solution to **pH 2.0** to **3.0** by adding dropwise **2N HCl**. The color should shift toward **purple.**

Test for esters

Unlike the acid anhydrides and acid halides, esters react with hydroxylamine to form hydroxamic acids only when the reaction is carried out in the alkaline solution. Lactones act like esters in this test.

Place about **0.50** mL of **1.0N** hydroxyl ammonium chloride in methanol in a **4.0** inch test tube and add about **30.0** mg of the compound. Now add, dropwise, a **2N** solution of **KOH** in methanol until the mixture is alkaline to litmus and then add **4.0** drops more **2N** solution of **KOH** in methanol. Heat the mixture just to boiling, then cool it, and add, dropwise and with shaking, **2NHCl** solution until the **pH** of the reaction mixture is approximately **3.0**. Add **1.0** drop of **10.0% $FeCl_3$** solution and note the color. A **bluish-red** or **reddish-blue color** is a positive test for esters.

Tests for active unsaturation (C=C, C≡C)

(i) ***Bromine in carbon tetrachloride test*:** Dissolve about **50.0** mg of the compound to be tested in **1.0** mL of **CCl_4** and add dropwise a **2.0%** solution of bromine in **CCl_4**. If more than **2.0** drops of the reagent are required to cause the bromine color to remain for at least **1.0** minute, a reaction is indicated. If hydrogen bromide is not produced **[determine by blowing across the top of the reaction tube and noting the fog that is produced, if HBr is produced]**, the compound is unsaturated. Alkynes react slowly with bromine.

(ii) ***Potassium permanganate test*:** Dissolve about **25.0** mg of the compound in **2.0** mL of water or acetone **(alcohol free)** in a **4.0** inch test tube. Add a **1.0%** aqueous solution of potassium permanganate drop by drop, with vigorous shaking of the test tube. If more than **1.0** drop of the potassium permanganate is reduced, the compound may be an unsaturated hydrocarbon.

Test for α-amino acids

(i) ***Ninhydrin test*:** Dissolve about **1.0** to **5.0** mg of the substance in about **1.0** to **3.0** mL of water in a **4.0** inch test tube. Add **1.0** mL of **0.10%** solution of ninhydrin in water. Heat the contents of the test tube to boiling for **1.0** to **3.0** minutes. The production of a **violet, blue** or **purple color** is a

positive test of amino acids. Other colors (yellow, orange, red) are negative test. α-Amino acids and β-amino acids react with ninhydrin to give positive test, which is a blue to blue-violet color.

(ii) ***Nitrous acid test*****:** Dissolve about **0.50** mg of the sample, α-amino acid in about **1.5** mL of acetic acid and dilute with about **2.5** mL of water, and cool the solution to **0-5°C** temperature in a beaker of ice. Dissolve about **0.50** mg of sodium nitrite in about **2.5** mL of water, add the solution dropwise, with shaking, to the cold solution of α-amino acid in acetic acid. Evolution of nitrogen gas indicates the positive test for α-amino acid. α-Amino acid reacts with the nitrous acid to form an intermediate, which decomposes to nitrogen gas and α-hydroxy acid.

(iii) ***Copper (II) sulfate test*****:** Dissolve a small amount of α-amino acid to be tested in **1.0** mL of water in a **6.0** inch test tube. Add **2.0** drops of **1.0M** copper **(II)** sulfate solution in the aqueous solution of α-amino acid. If a blue color is not formed immediately, then heat the test tube in a hot-water bath for **5.0** minutes. A moderate to deep-blue liquid or a dark-blue solid is a positive test for α-amino acid. The deep blue color is due to the formation of copper complex.

Test for carbohydrates

(i) ***Anthrone test*****:** Anthrone dissolved in concentrated H_2SO_4 will react with all but the most complex carbohydrates to produce a **green color.** This test is relatively specific for carbohydrates if the reaction mixture is not heated.

(ii) ***H_2SO_4 test*****:** In a dry test tube take about **0.10** g of the sample to be tested and add **2-3.0** drops of concentrated H_2SO_4. A **red** or **crimson color** indicates the presence of a glucoside.

(iii) ***p-Toluidine acetate test*****:** Place about **5-10.0** mg of the sample in a microcrucible. Cover the crucible with a filter paper that has been moistened with **2.0** drops of a **10.0%** solution of *p*-toluidine in **10.0%** acetic acid. Heat the bottom of the crucible for **1.0** minute. If a **pink** to **red color** does not form on the filter paper, repeat the experiment, adding a drop of syrupy phosphoric acid to the sample and heat it. Production of **red color** with *p*- toluidine acetate is a positive test for carbohydrate **[heat or hot phosphoric acid produces furfural or furfural derivatives that produce the color with *p*-toluidine acetate]**.

(iv) ***Molisch test*****:** Dissolve about **20.0** mg of the sample to be tested in **1.0** mL of water and add **2.0** drops of **5.0%** solution of α-naphthol in methanol. Place **1.0** mL of concentrated H_2SO_4 in a **4.0** inch test tube and while holding the tube at an angle, slowly introduce the solution to be tested by means of a pipette so that it stratifies on top of the acid. The development of a **violet-purple color** at the junction of two layers indicates the presence of carbohydrate.

Differentiating the monosaccharides from disaccharides

Barfoed's reagent test**:** Monosaccharides are easily oxidized than the disaccharides by Barfoed's reagent. Place about **2.0** mL of Barfoed's reagent in a **4.0** inch test tube and add about **10-20.0** mg of the sample to be tested, shake and place the test tube in a bath of boiling water for **2-3.0** minutes. Remove the test tube from the bath and allow it to cool. A **yellow orange** or **orange-red** precipitate is a positive test for monosaccharide **(glucose, fructose or galactose)**. If **red** precipitate is formed only after a prolonged heating then the presence of a disaccharide **(sucrose etc.)** is indicated. [**A solution of cupric acetate and acetic acid is called Barfoed's reagent**].

Distinguishing ketoses from aldoses

Seliwanoff's test**:** Mix about **1.0** mL of Seliwanoff's reagent with **1.0** mL of **5.0%** solution of the sugar to be tested in water. Heat the mixture to boiling. A **red color** develops within **2.0** minutes if the sugar is a ketose; on the other hand, long standing or prolonged heating will develop the color with aldoses. [**A solution of resorcinol and HCl or concentrated sulfuric acid is called Seliwanoff's reagent**].

Preparation of derivatives

A derivative may be defined as a substance derived or prepared from some other compound usually retaining the general skeleton of the original substance. Usually the functionalities of the original compound are changed, **e.g.,** acetanilide, is a derivative of aniline.

A suitable derivative should fulfill the following requirements

(i) It should be quickly and conveniently prepared.
(ii) It should be a solid preferably crystalline having **m. p.** between **50-250°C.**
(iii) It should be readily purified by crystallization.
(iv) It should be obtainable by reagents available in the laboratory.
(v) It should differ with the original compound in chemical and physical properties.
(vi) It should be obtainable in good yields within half an hour.
(vii) The process employed should be generally applicable to whole class of compound rather than an individual compound.

The preparation of a solid, crystalline derivative is the final stage in the identification of the organic compounds. The conformity of the **m. p.** of a particular derivative of the sample with the literature value is a conclusive proof of the unknown. A list of suitable derivatives of the following class of compounds is given below:

Compounds	**Derivatives**
Acids	: amide, anilide.
Alcohols	: Acetals, benzoates.
Phenols	: Acetates, *p*-tosylates, picrates, bromides.
Aldehydes, Ketones	: 2,4-dinitrophenolhydrazones, oximes.
Nitro compounds	: Polynitro and reduction to amines and then diazotization followed by coupling product.
Amino compounds	: Acetals, benzoates, picrates, acetyl.
Amides, anilides	: Hydrolysis to acids, nitro, picrates, bromo.
Hydrocarbons	: Picrates, oxidation of side chain and nitro.
Carbohydrates	: Acetates, oximes, osazones.
Sulphonic acids	: Sulphonamides.
Thiourea	: s-benzyl thiourea salt.

Procedure for derivatization

(i) ***2,4-DNP*** **:** Take about **5.0** mL of ***2,4***-dinitrophenylhydrazine solution in a dry test tube. Add **1.0** mL or **1.0** gm of the given sample and **1-2.0** drops of concentrated sulfuric acid. Shake the reaction mixture vigorously and warm for few minutes, cool and filter the precipitate thus formed and recrystallize it from alcohol, record the **m. p.** of the dry derivative.

(ii) ***Anilide derivative*****:** In a dry **10.0** mL quick fit flask take **1.0** gm of the acid and **2.0** mL of aniline. Fit the flask with an air condenser. Reflux the contents for half an hour, cool the reaction mixture, filter off the product and wash with cold water. Crystallize the product from alcohol, dry and determine the **m. p.**

(iii) ***Benzoate derivative*****:** Add **1.0** mL of the sample (alcohol) to **10.0** mL of acetone and **5.0** mL of benzyl chloride contained in a **100.0** mL flask. Add now **50.0** mL of aqueous **NaOH** solution

gradually with constant stirring till a precipitate is obtained. Pour all the contents in a beaker containing ice-water. Filter the product, wash with water, crystallize from alcohol and dry, determine the **m. p**.

(iv) ***Acetate derivative*****:** Take nearly **1.0** gm of the sample (phenol or alcohol) in a dry clean test tube; add **2.5** mL of acetic anhydride and **1.0** drop of concentrated sulphuric acid. Cork the test tube and shake vigorously for **5.0** minutes. Pour the contents to **15.0** mL cold water in a beaker. Stir the contents till the oil solidifies. Filter the product, wash with water and crystallize from hot water or dilute alcohol.

(v) ***Acetyl derivative*****:** In a small quick fit flask fitted with a reflux condenser take **1.0** gm or **1.0** mL of the amine, **2.0** mL of acetic anhydride and **2.0** mL of glacial acetic acid. Reflux for about **15.0** minutes. Pour the contents into a beaker containing water. Stir until the product solidifies. Filter and crystallize from dilute alcohol and dry. Determine the **m. p**. of the derivative.

(vi) ***Picrate derivative*****:** In a test tube prepare a saturated solution of the sample (amine, aromatic hydrocarbon) in alcohol or benzene. In another test tube prepare a saturated solution of picric acid in the same solvent. Mix the two solutions, shake and allow to stand for few minutes. If no solid separates, heat gently on a water-bath, cool, and filter off the picrate. Crystallize it from the least quantity of alcohol.

Dissolve **0.10** g of aromatic hydrocarbon and **0.20** g of picric acid separately in the minimum volume of hot rectified spirit **(**about **2.0 mL)**, mix the two solutions and allow to cool. Filter and wash with **2.0** mL ethanol. Recrystallize from hot ethanol, ethyl acetate or ether.

(vii) ***Bromo derivative*****:** Dissolve the compound in water/ alcohol/acetone or acetic acid and take the contents in a **100.0** mL flask. Add strong bromine solution dropwise **(**dissolve **1.5** g **KBr** in **10.0** mL of water and add **1.0** g of bromine**)** until, after shaking the liquid is pale yellow. Add **50.0** mL of water, cool and shake vigorously. Filter and wash the bromo derivative with water. Recrystallize the product from alcohol and dry. Determine the **m. p.**

(viii) ***Amide formation*****:** Add **0.50** g of the sample **(ester, lactone)** in **10.0** mL of concentrated ammonium hydroxide and shake well till a solid is separated. Filter, wash with water, crystallize from alcohol, dry and determine the **m. p.**